Lab 6: Metabolism - Enzymes, ATP, and Energy Flow

BIOL-1

Name: ______________________ **Date:** ______________________

Learning Objectives

By the end of this lab, you should be able to:

1. Explain how enzymes lower activation energy without being consumed.
2. Predict how temperature, pH, and inhibitors affect enzyme activity.
3. Model ATP as a short-term energy-transfer molecule.
4. Connect metabolism to photosynthesis and cellular respiration.

Overview

This lab uses a paper-and-pencil enzyme simulation to connect metabolism, ATP, and reaction rates. You will compare enzyme behavior under normal, hot, acidic, and inhibited conditions, then interpret why cells need controlled chemical pathways.

Materials Needed

- Pencil or pen
- Calculator
- This worksheet

Part 1: Enzyme Baseline

1. In one sentence, what does an enzyme do to activation energy?

2. A reaction without enzyme produces **2 products/minute**. With enzyme it produces **18 products/minute**. How many times faster is the enzyme condition?

Condition	Products after 1 min	Products after 2 min	Pr
Normal enzyme			
High heat			
Low pH			
Competitive inhibitor			

Part 2: Interpret the Pattern

3. Which condition caused the largest drop in enzyme activity? Use numbers from your table.

4. Explain why a denatured enzyme cannot simply keep working at the same rate.

5. ATP is often called cellular "energy currency." Explain what that analogy means and one way it can be misleading.

Part 3: Metabolism Map

Draw arrows connecting these ideas: **food energy**, **ATP**, **cell work**, **heat**, **photosynthesis**, and **cellular respiration**.

Exit Question

6. Why is metabolism not just "breaking down food," but a coordinated network of building and breaking reactions?